

$$(1) \quad v_{22} + 10.251$$

$$y'' + 4y' + 25y = \delta(t - \pi)$$

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$$y(0) = 1$$

$$y'(0) = 0$$

$$s^2 y - sy(0) - y'(0) + 4 \cdot 251 \cdot y = e^{-s\pi}$$

$$s^2 y - s + 4 \cdot 251^2 \cdot y = e^{-s\pi}$$

$$y(s^2 + 4 \cdot 251^2) = e^{-s\pi} + s$$

$$y = \frac{e^{-s\pi}}{s^2 + 4 \cdot 251^2} = \frac{e^{-s\pi}}{s^2 + (2 \cdot 251)^2}$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{e^{-s\pi}}{s^2 + 502^2} \right\} (t) + \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 502^2} \right\} (t)$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 502^2} \right\} (t - \pi) + \cos 502t \Rightarrow$$

$$\Rightarrow \frac{u(t - \pi)}{502} \cdot \mathcal{L}^{-1} \left\{ \frac{502}{s^2 + 502^2} \right\} (t - \pi) + \cos 502t \Rightarrow$$

$$\Rightarrow \frac{u(t - \pi) \cdot \sin(502(t - \pi))}{502} + \cos 502t.$$

$$y(t) = \frac{u(t - \pi) \cdot \sin(502(t - \pi))}{502} + \cos 502t.$$



$$② \quad f(t) = \begin{cases} 0, & 0 < t < 2.51 \\ \sin t, & 2.51 < t \end{cases}$$

$$f(t) = 0 \cdot \Pi_{0,2.51}(t) + \sin t \cdot \delta_{2.51}(t) \cdot u(t - 2.51)$$

$$f(t) = \sin t \cdot u(t - 2.51)$$

$$F(s) = \int_0^\infty \sin t \cdot u(t - 2.51) e^{-st} dt = e^{-s \cdot 2.51} \int_{2.51}^\infty \sin t e^{-s(t-2.51)} dt =$$

$$= e^{-2.51s} \cdot \int_0^\infty \sin(t+2.51) e^{-s(t+2.51)} dt =$$

$$= e^{-2.51s} \cdot \left[\frac{\cos 2.51s}{s^2+1} + \frac{s \cdot \sin 2.51s}{s^2+1} \right] = f(s)$$

$$③ \quad y'(t) + 4y(t) = f(t), \quad y(0) = 1, \quad y'(0) = 3$$

$$s^2 y + s y(0) + y'(0) + 4y = F(s)$$

$$s^2 y + s + 3 + 4y = F(s)$$

$$y(s^2 + 4) = F(s) + s + 3$$

$$y = \frac{F(s) + s + 3}{s^2 + 4}$$

$$② \quad y = \frac{F(s)}{s^2 + 4} + \frac{s}{s^2 + 4} + \frac{3}{s^2 + 4}$$

$$y(t) = \mathcal{L}^{-1} \left\{ \frac{F(s)}{s^2 + 4} \right\} (t) + \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 4} \right\} (t) + \mathcal{L}^{-1} \left\{ \frac{3}{s^2 + 4} \right\} (t) =$$

$$= \mathcal{L}^{-1} \{ F(s) \} (t) * \mathcal{L}^{-1} \left\{ \frac{1}{s^2 + 2^2} \right\} (t) + \cos 2t + \frac{3}{2} \sin 2t =$$

$$= \sin t \cdot u(t - 2.51) * \frac{1}{2} \sin 2t + \cos 2t + \frac{3}{2} \sin 2t =$$

$$= \int_0^t \sin(2t - 2v) f(v) dv + \cos 2t + \frac{3}{2} \sin 2t$$

$$\textcircled{3} \quad \frac{s}{(s^2 + 251^2)(s - 4)} = F(s)$$

$$f(s) = \mathcal{L}^{-1} \left\{ \frac{s}{(s^2 + 251^2)(s - 4)} \right\} (t) = \mathcal{L}^{-1} \left\{ \frac{s}{s^2 + 251^2} \right\} * \mathcal{L}^{-1} \left\{ \frac{1}{s - 4} \right\} (t) =$$

$$= (\cos 251t * e^{4t}) = \int_0^t \cos 251(t-v) \cdot e^{4v} dv =$$

$$= \int_0^t [\cos 251t + \cos 251v + \sin 251t \sin 251v] e^{4v} dv =$$

$$= \int_0^t e^{4v} \cdot \cos 251t \cdot \cos 251v + \int_0^t e^{4v} \sin 251t \cdot \sin 251v dv =$$

$$= \cos 251t \int_0^t \cos(251v) e^{4v} dv + \sin 251t \int_0^t \sin 251v e^{4v} dv =$$

$$= \frac{4 \cdot \cos 251t}{16 + 251^2} \left(\cos 251t e^{4t} + \frac{251}{4} \sin(251t) e^{4t} - 1 \right) +$$

$$+ \frac{4 \sin 251t}{16 + 251^2} \left(\sin 251t e^{4t} - \frac{251}{4} \cos 251t e^{4t} + \frac{251}{4} \right)$$



$$\begin{cases} X'' + 251^2 Y = 1 & X(0) = 1, X'(0) = 1, Y(0) = 1, Y'(0) = -1 \\ 251^2 X + Y'' = -1 \end{cases}$$

$$\begin{cases} S^2 X - S - 1 + 251^2 Y = \frac{1}{S} \Rightarrow S^2 X + 251^2 Y = \frac{S^2 + S + 1}{S} \\ 251^2 X + S^2 Y - S + 1 = -\frac{1}{S} \Rightarrow 251^2 X + S^2 Y = \frac{S^2 - S - 1}{S} \end{cases}$$

$$1) 251^2 Y = \frac{S^2 + S + 1}{S} - S^2 X \Rightarrow Y = \frac{S^2 + S + 1 - S^3 X}{S \cdot 251^2}$$

$$2) 251^2 X + S^2 Y = \frac{S^2 - S - 1}{S}$$

$$251^2 X + S^2 \left(\frac{S^2 + S + 1 - S^3 X}{S \cdot 251^2} \right) = \frac{S^2 - S - 1}{S}$$

$$251^2 X + \frac{S^3 + S^2 + S}{251^2} - \frac{S^4 X}{251^2} = \frac{S^2 - S - 1}{S}$$

$$X \left(251^2 - \frac{S^4}{251^2} \right) = \frac{S^2 - S - 1}{S} - \frac{S^3 + S^2 + S}{251^2}$$

$$X \left(\frac{251^4 - S^4}{251^2} \right) = \frac{251^2 S^2 - 251^2 S - 251^2 - S^4 - S^3 - S^2}{251^2 S}$$

$$X = \frac{251^2 S^2 - 251^2 S - 251^2 - S^4 - S^3 - S^2}{S(251^4 - S^4)}$$

$$X = \frac{S^2(251^2 - S^2) - S(251^2 + S^2) - (251^2 + S^2)}{S(251^4 - S^4)}$$

$$X = \frac{S}{251^2 + S^2} - \frac{1}{251^2 - S^2} - \frac{1}{S(251^2 - S^2)}$$

$$X = \frac{S}{251^2 + S^2} + \frac{1}{S^2 - 251^2} + \frac{1}{S(S^2 - 251^2)} = \left[\frac{1}{S(S^2 - 251^2)} = \frac{A}{S} + \frac{B}{S - 251} + \frac{C}{S + 251} \right] \Rightarrow$$

$$\Rightarrow X(t) = \frac{S}{251^2 + S^2} + \frac{1}{S^2 - 251^2} - \frac{1}{251^2 S} + \frac{1}{2 \cdot 251^2 (S - 251)} + \frac{1}{2 \cdot 251^2 (S + 251)}$$





$$x(t) = \cos 251t + \frac{1}{251} \sinh 251t - \frac{1}{251} + \frac{e^{251t}}{2 \cdot 251^2} + \frac{e^{-251t}}{2 \cdot 251^2}$$

$$x(t) = \cos 251t + \frac{1}{251} \sinh 251t + \frac{e^{251t}}{2 \cdot 251^2} + \frac{e^{-251t}}{2 \cdot 251^2} - \frac{1}{251^2}$$

$$1) 2s^2x + s^2y = \frac{s^2 - s - 1}{s}$$

$$x = \frac{s^2 - s - 1 - s^3y}{251^2s}$$

$$2) s^2x + 251^2y = \frac{s^2 + s + 1}{s}$$

$$s^2 \left(\frac{s^2 - s - 1 - s^3y}{251^2s} \right) + 251^2y = \frac{s^2 + s + 1}{s}$$

$$\left(\frac{s^4}{251^2} - 251^2 \right) y = \frac{s^3 - s^2 - s}{251^2} - \frac{s^2 + s + 1}{s}$$

$$\left(\frac{s^4 - 251^4}{251^2} \right) y = \frac{s^2(s^2 - 251^2) - s - (s^2 + 251^2) - (s^2 + 251^2)}{s \cdot 251^2}$$



$$y = \frac{s}{s^2 + 251^2} - \frac{1}{251} \cdot \frac{251}{s^2 - 251^2} - \frac{1}{s(s^2 - 251^2)}$$

$$y(t) = \sin 251t - \frac{1}{251} \sinh 251t + \left(\frac{1}{251^2s} - \frac{1}{2 \cdot 251^2(s - 251)} + \frac{1}{2 \cdot 251^2(s + 251)} \right)$$

$$y(t) = \sin 251t - \frac{1}{251} \sinh 251t - \frac{e^{251t}}{2 \cdot 251^2} + \frac{e^{-251t}}{2 \cdot 251^2} + \frac{1}{251^2}$$